

POC 02: Smart Building Management Platform - CI/CD Modernization

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Role: **AWS DevOps Engineer**

Public-safe name: **Smart Building Management Platform**

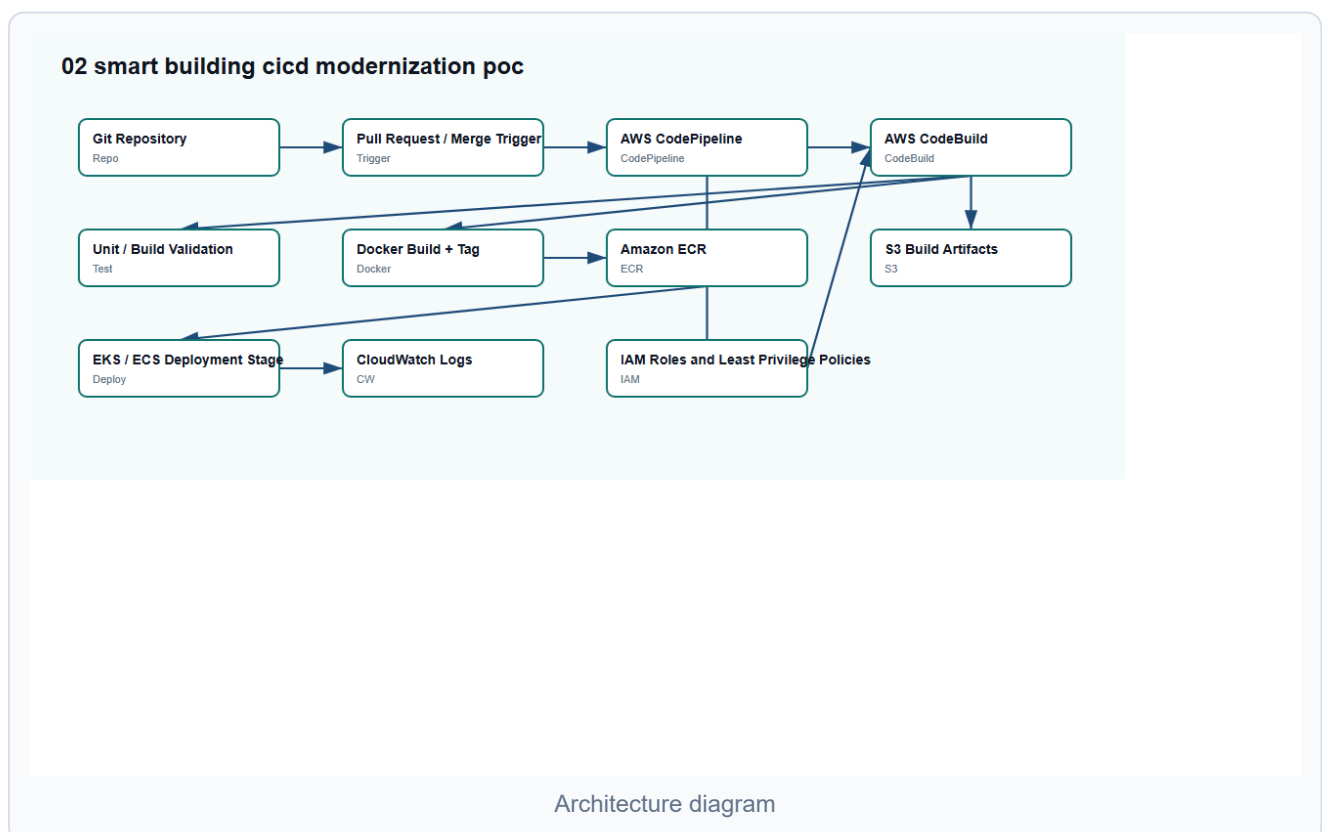
Confidentiality Note

This POC is public-safe and uses generic source, build, and deployment names. No private pipeline values, repository names, credentials, or client identifiers are included.

POC Objective

Design a migration pattern from Jenkins-heavy build workflows toward AWS-native CI/CD using CodePipeline, CodeBuild, Docker image builds, Amazon ECR, S3 artifact storage, IAM roles, CloudWatch logs, and deployment handover notes.

CI/CD Architecture Diagram



What I Worked On

- â€¢ Documented existing Jenkins release flow and migration considerations.
- â€¢ Planned CodePipeline and CodeBuild build stages.
- â€¢ Documented Docker image build, tagging, ECR push, S3 artifact handling, IAM permissions, and CloudWatch logging.
- â€¢ Prepared handover notes for release support and troubleshooting.

POC Implementation Scope

- â€¢ Dummy app repository with Dockerfile.
- â€¢ Buildspec file for CodeBuild.
- â€¢ CodePipeline stages for source, build, and deployment placeholder.
- â€¢ ECR image naming and tagging standard.
- â€¢ CloudWatch log group and retention notes.
- â€¢ IAM least-privilege checklist.

Suggested Repository Structure

```
aws-cicd-modernization-poc/  
|-- README.md  
|-- app/  
|   |-- Dockerfile  
|   `-- health.html  
|-- buildspec.yml  
|-- terraform/  
|   |-- codebuild.tf  
|   |-- codepipeline.tf  
|   |-- ecr.tf  
|   |-- iam.tf  
|   `-- s3.tf  
`-- docs/  
    |-- release-runbook.md  
    `-- troubleshooting.md
```

Validation Steps

- Run Docker build locally for the dummy app.
- Validate `buildspec.yml` commands.
- Run Terraform `fmt`, `validate`, and plan-only review.
- Confirm pipeline stages and IAM responsibilities are documented.
- Confirm CloudWatch log and artifact retention notes are present.

Expected Outcome

- â€¢ Clear Jenkins-to-AWS CI/CD migration blueprint.
- â€¢ Repeatable image build and ECR publishing pattern.
- â€¢ Better release visibility through CloudWatch logs and documented handover.

Technologies

Jenkins, AWS CodePipeline, AWS CodeBuild, Docker, Amazon ECR, Amazon S3, IAM, CloudWatch, Terraform, Kubernetes/ECS deployment concepts.